Sneha Vireshwar Dixit

sdixit2@huskers.unl.edu| snehavdixit.github.io

Education

PhD. Student : University of Nebraska-Lincoln — Aug 2023 - Present

Subject : Physics | Advisor : Dr. Ken Bloom | Chancellor's Fellowship Awardee

Bachelor of Science : Washington College — Jun 2020 - May 2023

Majors : Physics and Mathematics | Phi Beta Kappa, John S. Toll Physics Award

Research Experience

Effective Field Theory Analysis of Top Processes | University of Nebraska-Lincoln — Jan 2025 - Present

URA Visiting Scholars Award Fall 2026 Recipient | Advisor : Dr. Ken Bloom

- Wrote TopEFT processor for migration of analysis to Coffea 2024 hosted at UNL Tier-3.
- CMS Production and Reprocessing : Monitoring workflows and maintaining operations documentation.
- CMSDAS 2025 : Attended CMS Data Analysis School 2025 @ Fermilab LPC. Runners up in presentation competition for toy analysis of $t\bar{t}\gamma$ process.

IRIS-HEP Analysis Grand Challenge (AGC) | University of Nebraska-Lincoln — Jan 2024 - Aug 2024

- Tested and benchmarked Dask client performance in 2024 AGC pipeline as part of 200 gbps challenge.
- Created sample notebook for 2024 AGC pipeline with Dask and Coffea 2024 for ATLAS use.
- Links to : Pull request on Calver Coffea Demo, Presentation Slides

US ATLAS Researcher | Washington College — Oct 2022 - Jul 2023

US ATLAS SUPER 2023 grant recipient | Advisor : Dr. Suyog Shrestha

- Studied trigger efficiency in search for 0-lepton di-Higgs processes. Links to : Presentation Slides
- Set up new particle physics lab suited for computer-based physics analysis @ Washington College. Configured, benchmarked, and tested software and hardware for analysis.

Exploring WIMP Dark Matter | Washington College — Mar 2022 - May 2023

Cater Society of Junior Fellows, J.S. Toll Science Fellows grant recipient

- Calculated DM relic abundance with a freeze-out model of WIMP dark matter.
- Talk at AstroPhilly 2022 Symposium. Links to : Presentation Slides, Poster

USCMS PURSUE Intern | Fermilab LPC, University of Nebraska-Lincoln — May 2022 - Aug 2022

Advisors : Dr. Nick Smith, Dr. Oksana Shadura

- Created 2015 NanoAOD era for CMS OpenData and converted pre-2016 CMS OpenData to NanoAOD.
- Links to : GitHub Repository, Pull Request on CMSSW 10.6.X, Presentation Slides

Background Modeling for ATLAS Di-Higgs Search | Washington College — May 2021 - Jul 2021

J.S. Toll Science Fellows grant recipient

- Simulated and validated top-quark background signals for di-Higgs processes. Links to : Poster

Teaching Experience

Graduate Teaching Assistant | University of Nebraska-Lincoln — Aug 2023 - May 2025

- Led recitation and physics help sessions for undergraduate physics while taking graduate classes.

Physics and Mathematics Tutor | Washington College — Oct 2021 - May 2023

- Bridge course design : Designed a math bridge course to help students with weak math backgrounds adjust to intro-level physics courses.
- Tutoring : Tutored physics and mathematics at the Washington College Quantitative Skills Center.

Skills

Programming : Python, ROOT, C++, Wolfram Mathematica, Unix+Bash/Shell, Virtual Machines with Docker, Git+Github, Coffea , HTML/CSS, Android Studio.
Research : Literature review, scientific modeling, grant + report writing, data collection + visualization, statistical analysis.
Lab : Circuit design + implementation, bread/protoboarding, oscilloscope, soldering.

Outreach

STEM Sisters @ Washington College — Feb 2022 - May 2023

- Founded interest group for women+gender minorities in STEM at Washington College.
- Conducted weekly python, HTML, and tech literacy classes for women+gender minority students.

Particle Physics Outreach in Bengaluru, India — Jan 2023

Washington College | Cater Society of Junior Fellows grant recipient

- Extended PURSUE 2022 research to create particle physics outreach program for high school students in Bengaluru, India. Used python instead of ROOT and Linux for accessibility.
- Students analyzed Z boson decay to muon-antimuon pairs and plotted resonance using CMS OpenData and python.
- Links to : Lecture Slides (1), Lecture Slides (2), Project code+plots generated by students

Improving Science Education @ Public Schools in Bengaluru, India — Jul 2022 - Oct 2022

- Worked with non-profit Curiouscity to improve science education in public schools in Bengaluru, India.
- Designed cost-effective experiments that can be repeated by students at home to explore scientific concepts.
- Made accessible and lucid educational posters for scientific literacy at school.